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AFNLP

PEELing Argumentation

Scaffolding Elementary Argumentative Writing with Decomposed LLM Feedback

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MOTIVATION

Argumentative writing depends on feedback that classrooms cannot supply

01

One-on-one feedback

Effective argumentation pedagogy relies on intensive individual scaffolding from a teacher.

02

Large class sizes

Frequency and quality of personalised guidance drop as the teacher's load rises.

03

"One-shot" submission

Comments arrive after grading, leaving no opportunity for iterative refinement.

MOTIVATION

Primary students already argue — the paragraph is where they break down

WHY IT MATTERS EARLY

Argumentation underpins academic and professional work across disciplines, from law to business negotiation.

Strengthening it in childhood improves critical thinking, logical reasoning and communication strategy.

WHAT P5–P6 STUDENTS FACE

Argument writing is already in the curriculum, and debate competitions are increasingly common.

Yet students struggle to **expand a key point into a well-structured paragraph** .

Direct instruction and strategy modelling — not minimal guidance — drive upper-primary writing development.

WHAT P5–P6 STUDENTS NEED

Granular, **decomposed** scaffolding they can act on

Tangible, **visual** representations of abstract structure

Pedagogical framing that prevents **passive outsourcing**

The target skill is narrow and teachable: turning one claim into Point, Evidence, Explanation and Link.

RESEARCH QUESTIONS

Three questions about skill, process, and experience

RQ1 To what extent does the PEEL-guided system improve students' **argumentative writing skills**?

RQ2 How does it influence students' **writing processes** — revision depth, length, feedback uptake?

RQ3 How does it affect **engagement, self-efficacy, and satisfaction**?

PEDAGOGICAL FOUNDATION

PEEL breaks one paragraph into four teachable moves

Point

State a clear claim that answers the question.

Already mastered by most Grade 5 writers.

Evidence

Support the claim with convincing reasons.

Requires selecting relevant support.

Explanation

Give facts, anecdotes or examples that justify the reason.

The most cognitively demanding move.

Link

Connect the argument back to the main idea.

Summarise, not repeat.

COGNITIVE LOAD THEORY Constraining an ill-defined task into four concrete components reduces extraneous load.

SYSTEM ARCHITECTURE

Four interconnected modules

EXPERT MODEL

Domain knowledge

Operationalises PEEL as the structural schema for decomposing every task and rubric.

STUDENT MODEL

Learner profile

Monitors input in real time and maps each sentence onto a PEEL component.

TUTORING MODEL

Feedback engine

Generates topics and immediate component-level scaffolding via LLMs.

USER INTERFACE

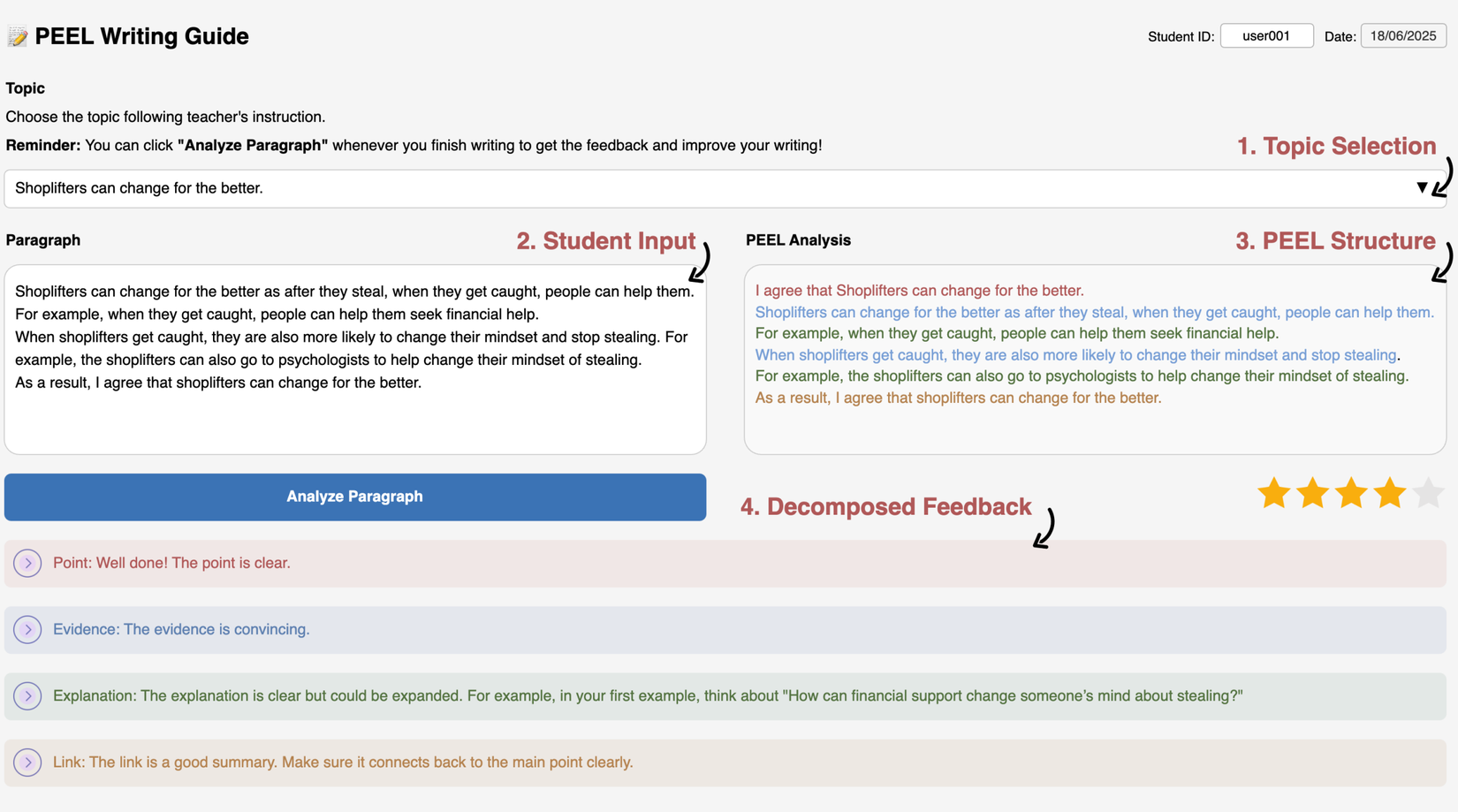
Colour-coded workspace

Age-appropriate layout that makes rhetorical structure visible.

INTERFACE

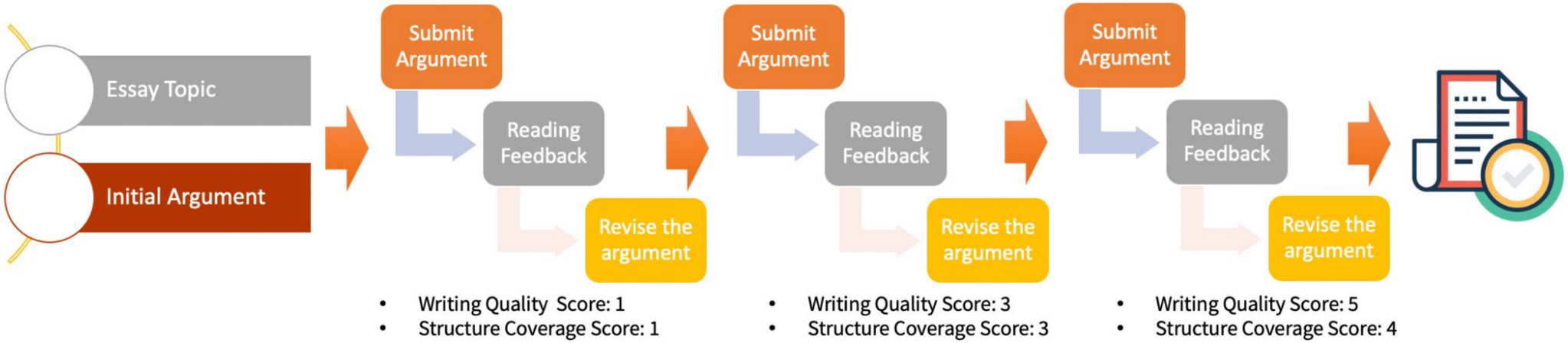
Writing, analysis and feedback on one screen

1 Topic 2 Paragraph 3 PEEL analysis 4 Component feedback



INTERACTION DESIGN

Writing as an iterative cycle, not a single submission



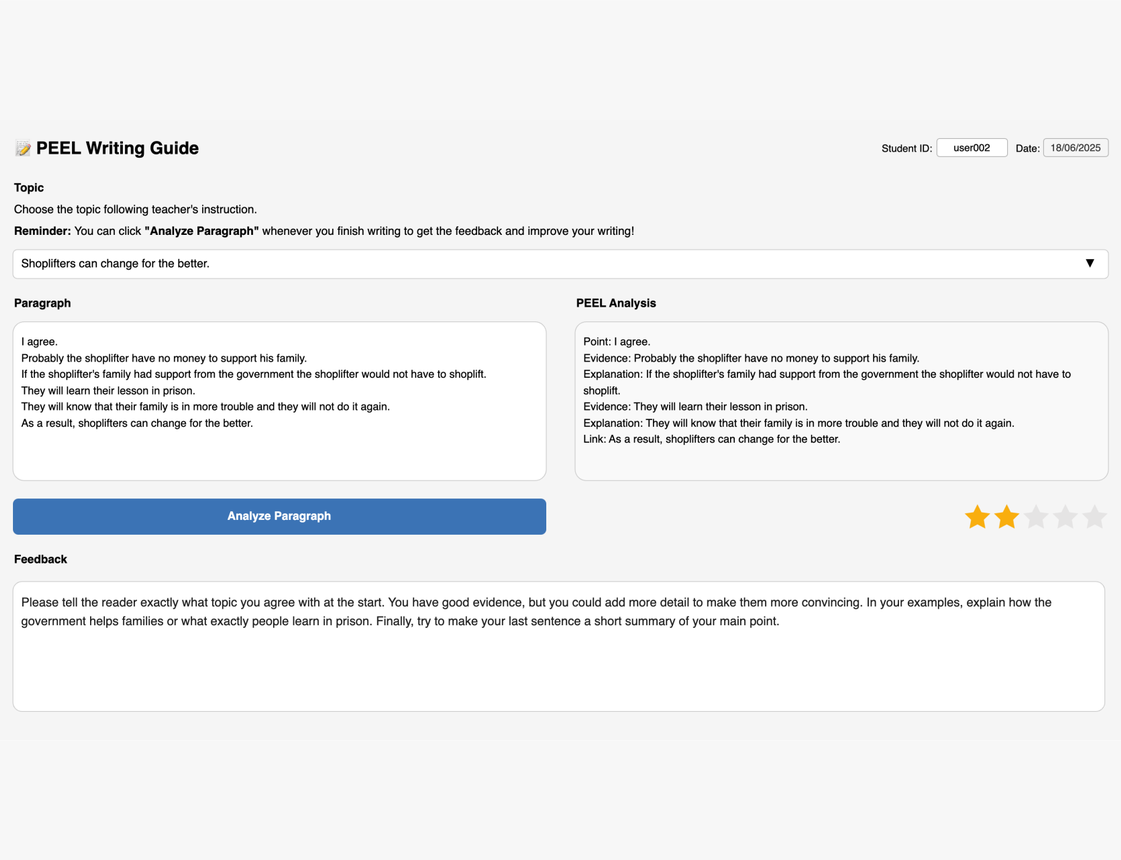
Each pass returns a structure-coverage and a writing-quality score, so the student can see the paragraph improve within a single 25-minute lesson.

COMPARISON CONDITIONS

Same model, same task — only the framing differs

A · Writing System_base

One unstructured plain-text feedback block, no PEEL decomposition.



PEEL Writing Guide

Student ID: user002 Date: 18/06/2025

Topic

Choose the topic following teacher's instruction.

Reminder: You can click **"Analyze Paragraph"** whenever you finish writing to get the feedback and improve your writing!

Shoplifters can change for the better. ▼

Paragraph

I agree.
Probably the shoplifter have no money to support his family.
If the shoplifter's family had support from the government the shoplifter would not have to shoplift.
They will learn their lesson in prison.
They will know that their family is in more trouble and they will not do it again.
As a result, shoplifters can change for the better.

PEEL Analysis

Point: I agree.
Evidence: Probably the shoplifter have no money to support his family.
Explanation: If the shoplifter's family had support from the government the shoplifter would not have to shoplift.
Evidence: They will learn their lesson in prison.
Explanation: They will know that their family is in more trouble and they will not do it again.
Link: As a result, shoplifters can change for the better.

Analyze Paragraph

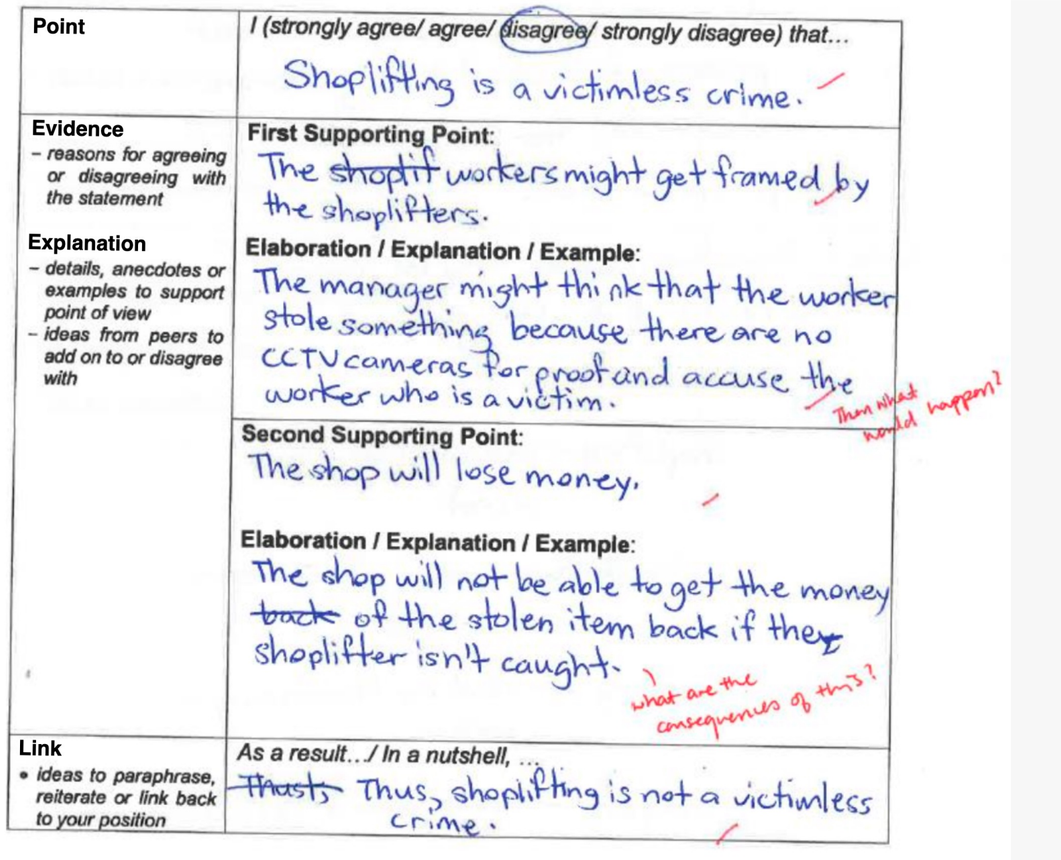
★★☆☆☆

Feedback

Please tell the reader exactly what topic you agree with at the start. You have good evidence, but you could add more detail to make them more convincing. In your examples, explain how the government helps families or what exactly people learn in prison. Finally, try to make your last sentence a short summary of your main point.

B · Teacher-led control

Paper worksheet; teacher comments returned in the next lesson.



Point	*I (strongly agree/ agree/ disagree/ strongly disagree) that…* Shoplifting is a victimless crime.
Evidence – *reasons for agreeing or disagreeing with the statement* **Explanation** – *details, anecdotes or examples to support point of view* – *ideas from peers to add on to or disagree with*	**First Supporting Point:** The ~~shoplif~~ workers might get framed by the shoplifters. **Elaboration / Explanation / Example:** The manager might think that the worker stole something because there are no CCTV cameras for proof and accuse the worker who is a victim. Then what would happen?
	Second Supporting Point: The shop will lose money. **Elaboration / Explanation / Example:** The shop will not be able to get the money ~~back~~ of the stolen item back if the shoplifter isn't caught. what are the consequences of this?
Link • *ideas to paraphrase, reiterate or link back to your position*	*As a result… / In a nutshell, …* ~~Thusly~~ Thus, shoplifting is not a victimless crime.

METHOD

A four-week classroom experiment with 130+ fifth-graders

EXPERIMENTAL 2	EXPERIMENTAL 1	CONTROL
PEEL-guided system	**Writing System_base**	**Teacher-led, paper-based**
n = 52	**n = 39**	**n = 39**

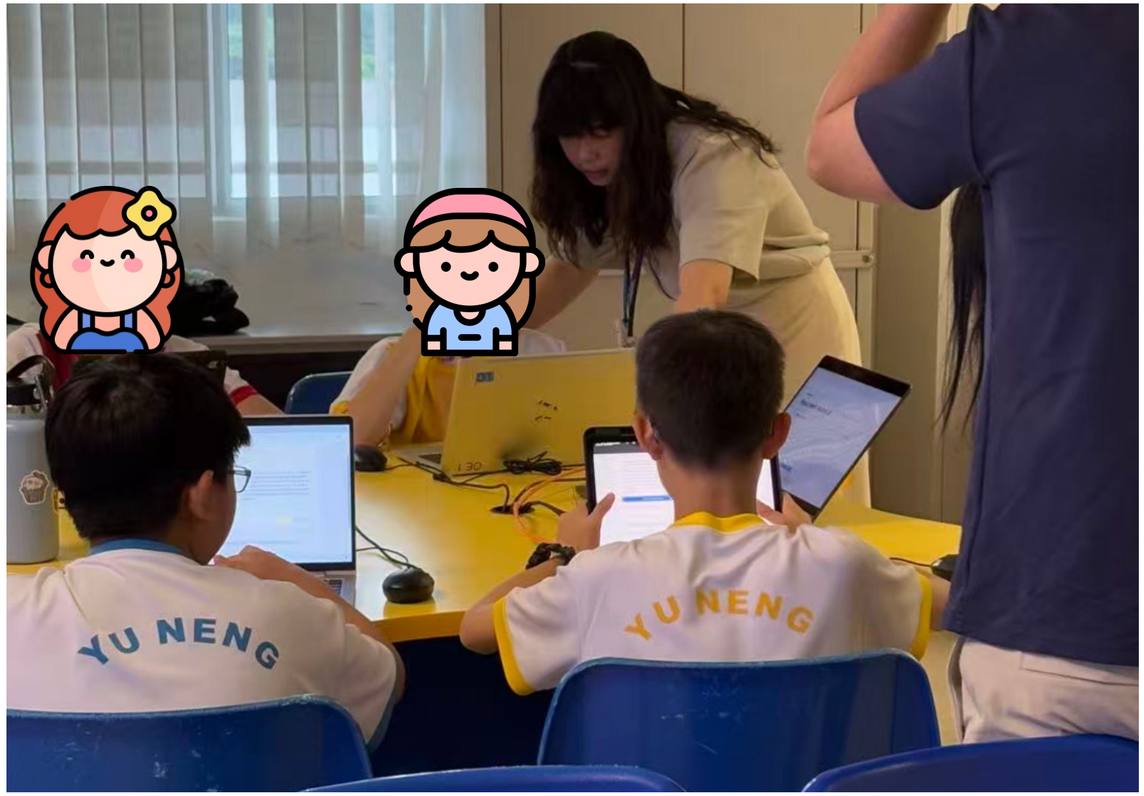
WEEK 0	WEEKS 1–4	WEEK 4
Pre-test + survey	Weekly 25-minute argumentative paragraph in regular English lessons	Post-test + survey

Measures: writing rubric, self-efficacy, engagement (α = .82/.90/.79), satisfaction (α = .90) Baseline equivalence: F(2,127) = 0.74, p = .481

DEPLOYMENT

Inside a regular English lesson, not a lab





15 MIN

Teacher instruction

25 MIN

Individual writing practice

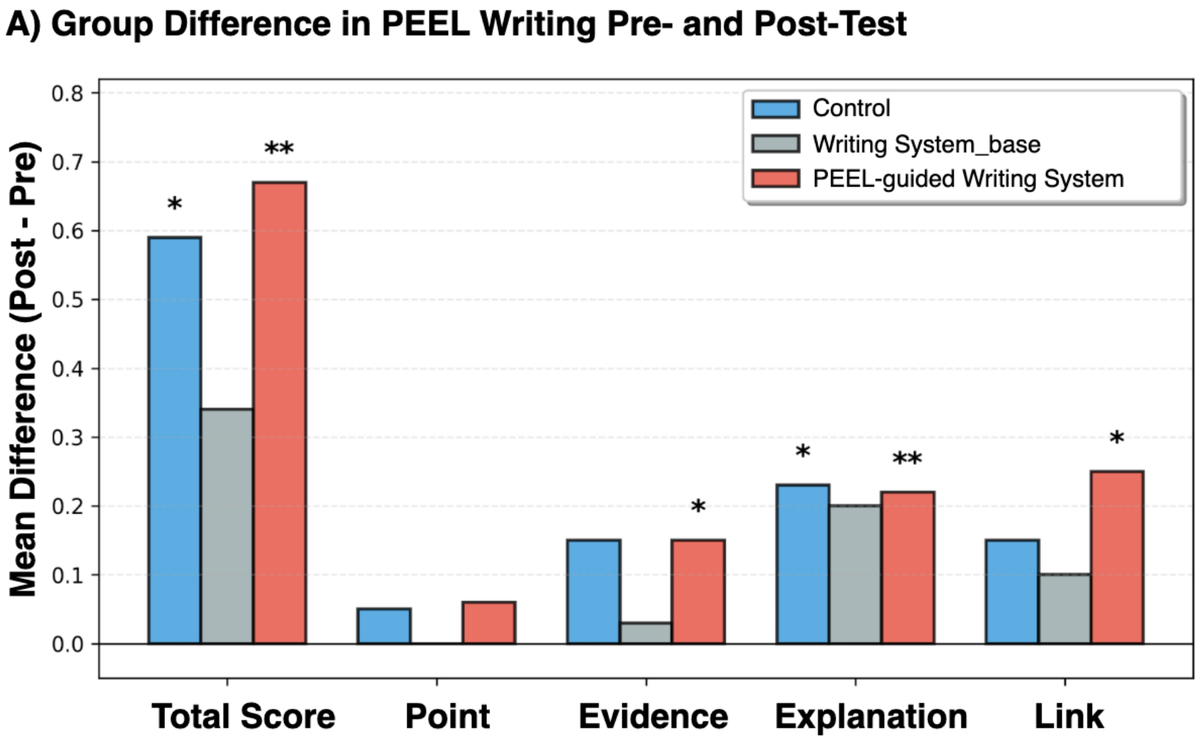
10 MIN

Sharing & reflection

Yu Neng Primary School · MOE and IRB approved · pre/post tests taken without AI

RQ1 · WRITING PERFORMANCE

Only structured scaffolding produced gains in Evidence and Link



PEEL-guided

M_{diff} = 0.67, d = 0.57, p < .01

Control

M_{diff} = 0.59, d = 0.41, p = .02

System_base

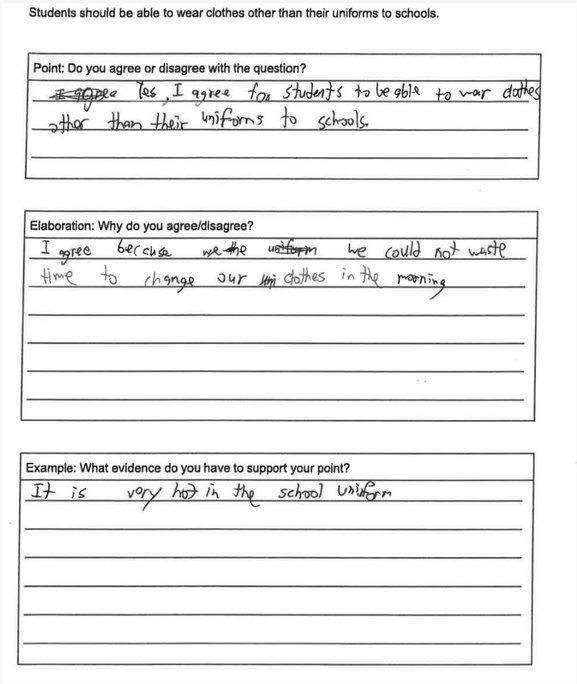
M_{diff} = 0.34, d = 0.19, **n.s.**

Point showed no change in any group (all p > .47) — Grade 5 writers can already state a claim.

RQ1 · WRITTEN WORK

Pre- and post-test writing samples

Pre-test · Week 0



Students should be able to wear clothes other than their uniforms to schools.

Point: Do you agree or disagree with the question?

~~Disagree~~ Yes, I agree for students to be able to wear clothes other than their uniforms to schools.

Elaboration: Why do you agree/disagree?

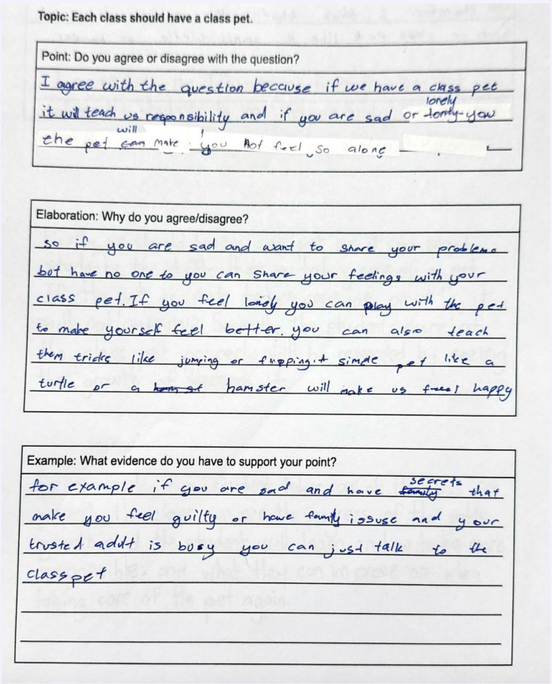
I agree because ~~we the uniform~~ we could not waste time to change our ~~my~~ clothes in the morning

Example: What evidence do you have to support your point?

It is very hot in the school uniform

One reason, one clause of evidence, no elaboration.

Post-test · Week 4



Topic: Each class should have a class pet.

Point: Do you agree or disagree with the question?

I agree with the question because if we have a class pet it will teach us responsibility and if you are sad or lonely you the pet will can make you not feel so along

Elaboration: Why do you agree/disagree?

so if you are sad and want to share your problems but have no one to you can share your feelings with your class pet. If you feel lonely you can play with the pet to make yourself feel better. you can also teach them tricks like jumping or flapping. A simple pet like a turtle or a ~~hams~~ hamster will make us feel happy

Example: What evidence do you have to support your point?

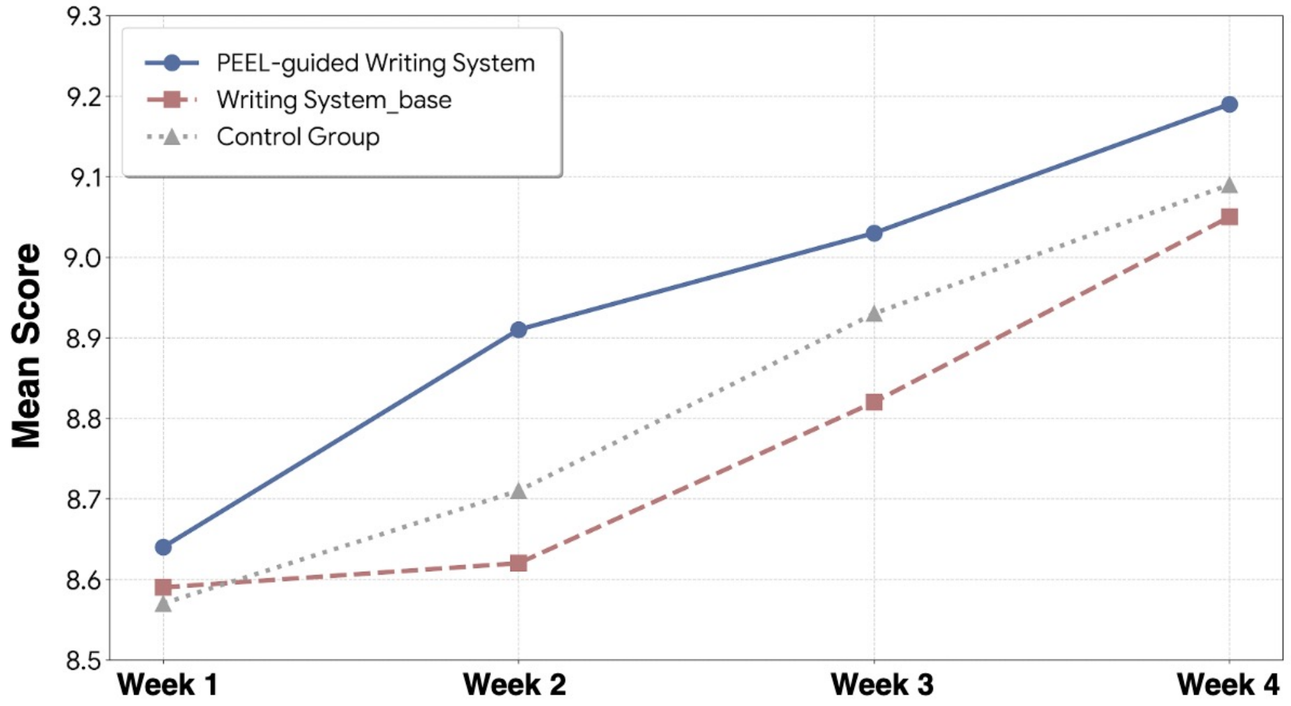
for example if you are sad and have ~~family~~ secrets that make you feel guilty or have family issues and your trusted adult is busy you can just talk to the classpet

Position plus two reasons, developed elaboration, concrete example.

Illustrative worksheets from the study — different prompts and different students, not a matched pair.

RQ1 · WEEKLY TRAJECTORY

The gap opens in week one and holds for four weeks



5.58

revision attempts per topic in the PEEL-guided group

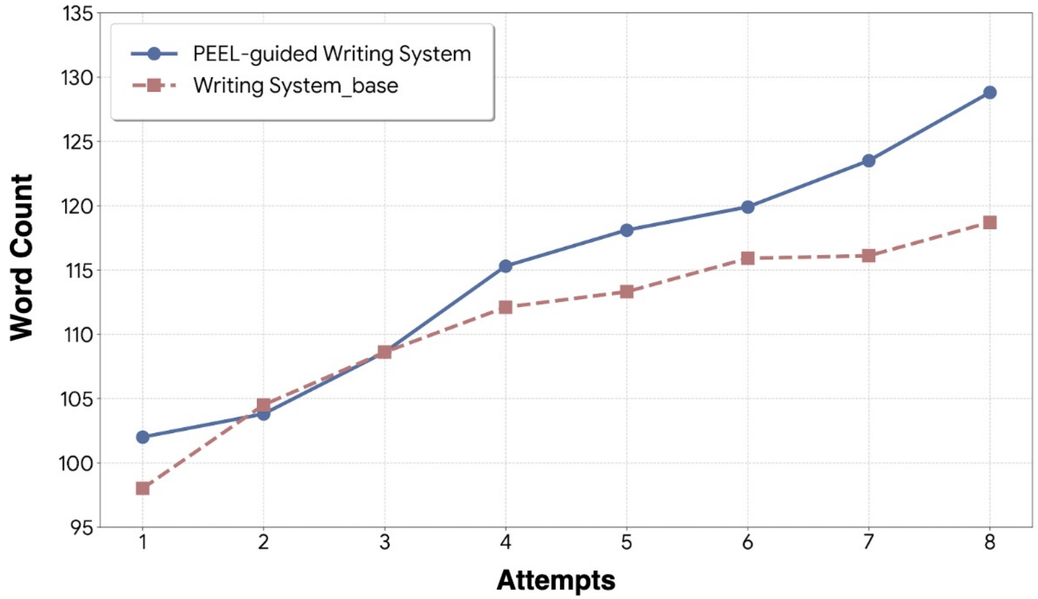
3.92

attempts in the unstructured feedback group

RQ2 · WRITING PROCESS

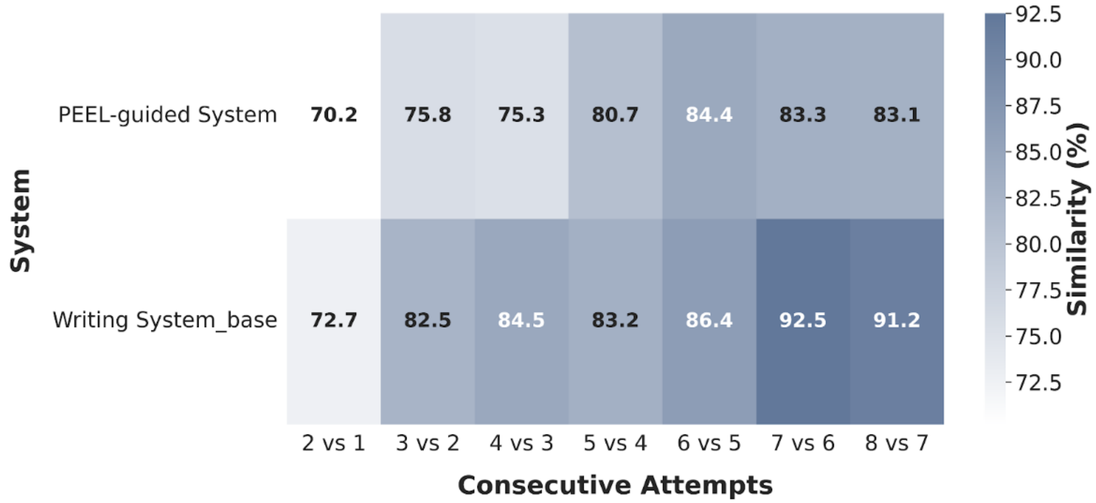
Decomposed feedback sustains revision; unstructured feedback stalls

Word count per attempt



102 → 128.8 words (**+26.3%**) vs 98 → 118.7 (+21.1%), plateauing after attempt 4.

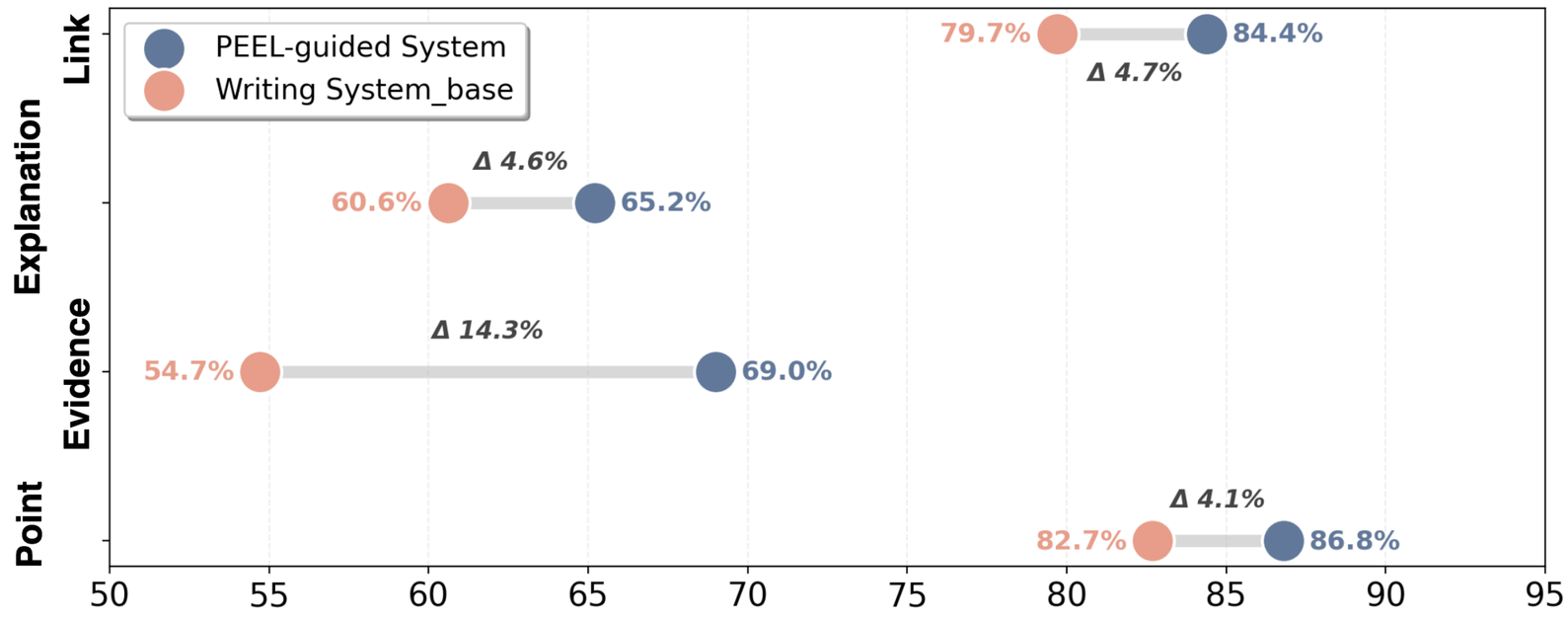
Similarity between consecutive attempts



Base-group edits shrink to **91–92%** similarity by attempt 7; PEEL holds **16–17%** change per round.

RQ2 · FEEDBACK UPTAKE

Adoption rises most where writing is hardest



Δ 14.3% higher adoption of **Evidence** feedback in the PEEL-guided system — the largest gap of the four components.

RQ2 · FEEDBACK THEMES

Both systems say much the same thing

FEEDBACK THEME	PEEL	BASE
Explanation: give example	**23.0%**	23.8%
Explanation: be specific	**13.6%**	11.1%
Evidence: clarify	**20.2%**	19.8%
Evidence: explain why	**20.1%**	21.9%
Link: connect back	**13.6%**	13.1%
Point: provide main idea	**5.7%**	5.9%
Social-emotional support	**3.8%**	4.3%

11,862

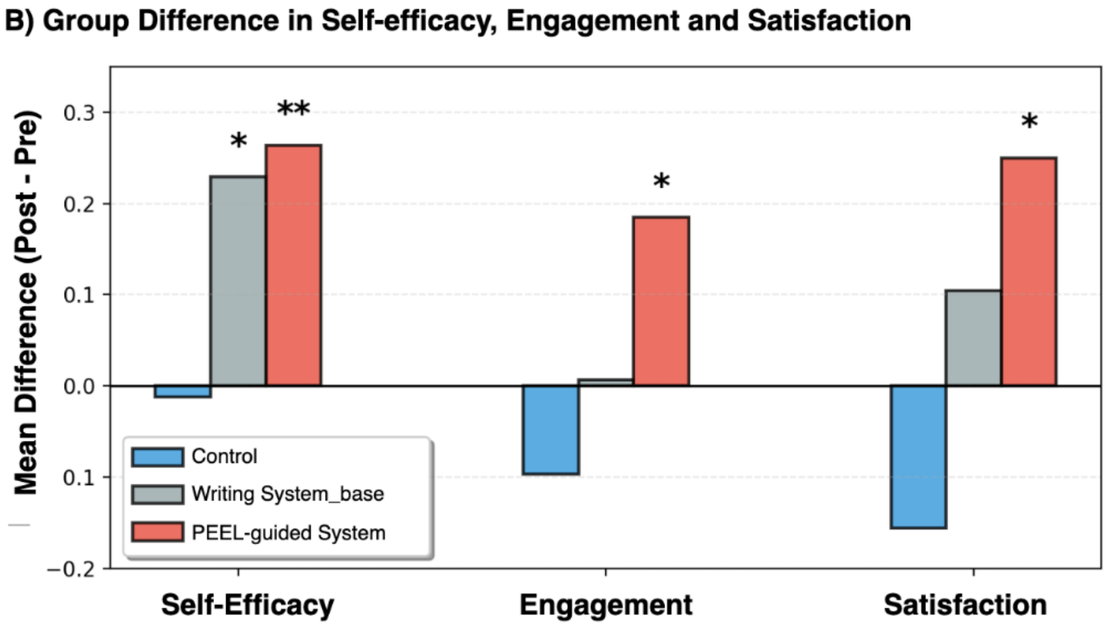
feedback instances in the PEEL-guided system, against 8,626 in the base system

Evidence and Explanation absorb three-quarters of all feedback in both conditions — the outcome difference comes from how it was framed, not what it said.

Both systems produced social-emotional encouragement without being prompted to.

RQ3 · AFFECTIVE OUTCOMES

Confidence rose in both AI groups; engagement only with PEEL



Self-efficacy	+0.26, d = 0.39, p = .002
Engagement	+0.19, d = 0.26, p = .03
Satisfaction	+0.25, d = 0.27, p = .03

47.3%
found it helpful

32.6%
found it fun

4.7%
saw score inconsistency

STUDENT VOICE

What the children said

> *"The AI was nice because it told me what to add or what was wrong."*
>
> Perceived utility · 47.3% of responses

> *"Super fun to write my work and let the robot look at it."*
>
> Enjoyment and engagement · 32.6%

> *"I was sad because I had 4 stars but the robot made me get less."*
>
> Scoring inconsistency · 4.7%

DISCUSSION

Why pedagogical framing, not model capability, made the difference

Cognitive load

Unstructured feedback asks a ten-year-old to interpret holistic comments, diagnose weaknesses, and plan a revision simultaneously.

PEEL constrains the task to four concrete components, redirecting cognitive resources to skill building.

Competence satisfaction

The control condition improved writing but ran on a passive submit-and-wait cycle with no visible progress.

Colour-coded analysis and star ratings show exactly which component is mastered, and let students act on it in the same lesson.

Effectiveness depended less on the underlying model than on the pedagogical constraints built around it.

CONCLUSION

Three takeaways

1 Framework-aligned scaffolding beat unstructured LLM feedback on writing quality; the same model without framing produced no significant gain.

2 Decomposition changed the writing process — longer drafts, deeper revisions, higher feedback adoption — not just the final score.

3 Only the PEEL condition improved engagement and satisfaction, pointing to a scalable model for age-appropriate AI tutors.

Thank you for your attention! Questions are welcome!